

APPENDIX 1 - DEFINITION OF TERMS

Globalization

Globalization encompasses processes of cross-border integration and interdependence facilitated by revolutions in technology, transportation, and communication (Held et al. 2000; Giddens 1997). Contemporary globalization is characterized by institutionalization through the Bretton Woods institutions, the United Nations, regional trade blocs, and multinational corporations (Dicken 2015). Sociological interpretations vary: some define globalization through social relations (Ritzer 2021; Robertson and White 2008), while others emphasize complex and contradictory forces (Browning, Halcli, and Webster 2000). Earlier phases were marked by Western imperialism, mercantilism, and industrialization (Robertson and White 2008). Some scholars identify a fourth phase beginning around the 2008 financial crisis, characterized by slowing growth, rising economic nationalism, and reversals of political integration (Axford 2021). This dissertation situates digital sovereignty within these contested global dynamics.

Digital globalization

Digital globalization describes the interplay of technology with forces of globalization as they reshape geopolitical, economic, cultural, and ecological relations. Revolutionary technological innovations, global digital networks, neoliberal ideology, and the power of transnational corporations distinguish the current phase (Sassen 2007; Tomlinson 2008). Some scholars posit a new global age marked by time-space compression and deterritorialization (Scholte 2005), while others argue contemporary patterns represent an intensification of earlier processes (Hopper 2007). Practically no domain of social life remains untouched. Major technology companies disproportionately influence this process through technical expertise, financial resources, and near-monopoly status. Their practices have detrimental social impacts that exemplify new matrices of power in contemporary

society (Couldry and Mejias 2019). This dissertation examines community-led responses to these concentrated digital power structures.

Agency and structure

A frequent sociological debate concerns the relationship between human agency and global structural forces (Sewell 1992). Some theorists emphasize how globalization constrains local agency, while others reveal creative local adaptations that reappropriate cultural flows (Appadurai 1996). Emirbayer and Mische (1998) conceptualize agency as engagement with structural environments, encompassing iterative, projective, and practical-evaluative dimensions. Sewell (1992) suggests agency and structure are inseparable. This dissertation addresses the agency-structure tension through the Design-Affordance-Sovereignty schema, where design choices (agency) operate within and reshape structural conditions, and affordances mediate between the two.

The role of the state

Sociologists investigate how globalization reshapes the role and sovereignty of the nation-state through multiple perspectives (Sassen 2007). World-systems theory argues states serve capitalist interests, allying to maintain hegemonic equilibrium (Wallerstein 2000), while others note how states strategically adapt to global pressures (Weiss 1998). States face deterritorialized threats from transnational actors, requiring new forms of governance (Scholte 2005). Neoliberal ideology has aligned states closer to market logics, constraining their autonomy and widening global inequalities (Brown 2019; Robinson 2004). Postcolonial theorists examine ongoing Northern domination, though some note how Southern states exercise more agency than assumed (Portes 1997; Weiss 1998). Chapter 5 treats the state as one of five sovereignty claimants alongside corporations, communities, international bodies, and technical institutions.

Market logic

Market logic requires the commodification of social life into a resource available for exchange (Polanyi 1957; Couldry and Mejias 2019). Drawing on Polanyi, Couldry and Mejias assert that market transformation requires a social transformation based on commodification of sociality. This dissertation documents how community-led digital initiatives construct alternative economic logics that resist market-driven extraction of community resources.

The fallacy of the ‘self-regulating’ market

Central to Polanyi's critique is the concept of fictitious commodification of land, labor, and money, a process he argues is foundationally misguided (Polanyi 1957). Commodification tore through the social fabric by dispossessing the poor in favor of the emerging market economy. The notional unlimited potential of material commodities created the belief that all human problems could be resolved, including those caused by commodification itself. Polanyi's argument implies that faith in the market as a naturally occurring order is also faith in modernity and technological progress through the commodification of labor. This faith creates a hierarchy where the market economy, a paradoxical and artificial invention, dominates human interests (Polanyi 1957). The dissertation's cases demonstrate alternatives to this logic.

Polanyi's concept of fictitious commodities as it relates to data

Michael Burawoy extended Polanyi's framework by identifying a fourth fictitious commodity: knowledge, not meant to be commodified for private interests but transformed by technologies into a sellable product through processes of dispossession (Burawoy 2015). Gernot Grabher contests this framing, arguing that knowledge and data are not fictitious commodities because they are not scarce and their marginal cost of reproduction is close to zero (Grabher 2021). Grabher documents the rise of data brokers, an industry exceeding \$1.5

trillion in annual sales that operates without meaningful oversight, packaging voice, transactions, browsing, geolocations, and consumption patterns into prediction products (Grabher 2021). As Pasquale argues, consumer protection and privacy laws are futile when companies can predict our health status without our health records (through purchasing, browsing, geographic and other data) and sell to us (Pasquale 2014).

This dissertation engages this debate through Research Question II, which examines whether community-led stewardship of digital resources generates collective wealth rather than extracted value.

Data Sovereignty

The principle that communities hold jurisdiction over data concerning their peoples, territories, and knowledge systems. Indigenous data sovereignty movements established foundational frameworks asserting that data governance is inseparable from self-determination, challenging extractive research practices through alternative protocols grounded in collective rights and cultural values (Carroll et al. 2020; Walter et al. 2021; Hudson et al. 2023). The CARE Principles for Indigenous Data Governance formalized these commitments into actionable standards. This dissertation operationalized data sovereignty as democratic practice DP2, one of nine outcome categories through which community-led digital initiatives constructed collective capacity. Across the 43 cases, data sovereignty functioned as both a specific governance achievement and a constitutive element of the broader sovereignty stack analyzed in Chapter 9.

Digital sociology

Digital sociology explores the inter-subjective relationship between digital technologies and society. Technologies mediate social relations in unpredictable ways as they embed themselves deeper into daily life, while designers and users imprint their values on the technologies themselves. The digital world contains forces of intensification oriented toward

influence, attention, and profit. These activities are instrumental rather than social, serving economic or political purposes (Marres 2017). Knowledge production through digital networks is mechanical and functions on code, creating new ontologies based on usage patterns and other data to categorize people. This technology-directed instrumentalization of sociality unravels the logics of sociality by making it machinic (Marres 2017). The dissertation's D-A-S schema addresses this instrumentalization through design analysis.

Decolonial sociology

Decolonial sociology draws from the ethic of pluriversality, where no single framework can offer universality. Instead, decolonial sociology involves multiple frameworks in conversation with each other on equal footing, identifying blind spots, challenging assumptions, and enriching a composite critical paradigm (Meghji 2021). This dissertation adopts a decolonial standpoint (Chapter 3) and advances a pluriversal digital sovereignty framework that centers community-based knowledge systems.

Decolonizing the digital

Decolonizing the digital refers to revitalizing the social, political, and economic relations embedded in and shaped by digital technologies to be more inclusive and democratic on a global scale. Decolonizing technology requires acknowledging its colonial roots, making values and biases explicit, centering marginalized groups, celebrating localized and contextual innovation, and building alternatives to data commodification. These require inclusive technology practices, participatory oversight and accountability, and transparency in algorithms and technological design. The 43 cases examined in this dissertation represent varied approaches to this decolonizing project.

Deterritorialization

Deterritorialization describes the weakening of territorial boundaries and the unmooring of activities from their geographic context (Appadurai 1990). It challenges fixed

notions of identity and community associated with the nation-state (Basch et al. 2005). Digital systems intensify deterritorialization by enabling governance, production, and collective action to operate across jurisdictions, complicating traditional sovereignty conceptions.

Imagined communities

Imagined communities are transnational identities and networks sustained despite geographic separation (Anderson 2016). They emerge from imagined bonds that replace eroding local ties (Appadurai 1990). Diasporic consciousness develops from displacement and global migration flows, with diaspora studies researching the connections and activities that bridge home and host societies (Basch et al. 2005; Glick Schiller, Basch, and Blanc-Szanton 1992). Chapters 5 and 8 extend this concept by examining how digital rights projects produce community through relational practices.

Transnational social fields

Transnational social fields denote spaces where relationships, exchanges, and meanings are constructed across national borders (Levitt and Schiller 2004). These fields contribute to the definition of the Digital Rights Project in Chapter 8, where relational ecosystems connecting cases, partners, movements, and networks operate across territorial boundaries.

The public sphere

The public sphere comprises interconnected networks between state, market, and civil society, engaging common issues (Emirbayer and Sheller 1999). Chapter 8 engages Tarkowski and Keller's (2021) vision of a "digital public sphere" where communities exercise collective authority over technological systems, connecting the concept to digital rights projects.

Democratic experiments

Democratic experiments are high-energy democracies with a strong democratic culture that is not servile to the dominant market order. They promote political expressions of difference, resolve impasse while minimizing compromise, exhibit devolution, and foster engagement in political life (Unger 2019). This dissertation adopts democratic experiments as an interpretive frame for the 43 community-led digital initiatives, analyzing the conditions under which they achieve democratic practices.

Inclusive vanguardism

Roberto Unger's inclusive vanguardism promotes economic, political, and psychological freedom through values of imagination and cooperation (Unger 2019). His prescribed approach covers education in generic, flexible, high-order capabilities; a paradigm of cooperation that disrupts the universality of property rights, contractual relations, and market organization; a renewal of the relationship between market and state; and experimentation with fragmented, conditional claims on the means of production. This includes intellectual property rights on data and revisiting relationships between platforms and customers. Inclusive vanguardism distinguishes itself from the current winner-takes-all situation where ownership is exclusive and power controls prices and access rights (Unger 2019). The dissertation draws on this concept across Chapters 2, 5, 7, and 8 as a normative tool for evaluating community-led digital initiatives.

Decolonial standpoint on technology and modernity

Giddens (1997), Beck (1992), and Castells (2010) argue that contemporary globalization represents the culmination of the project of modernity, compressing and universalizing certain values. The Western conception of modernity, which is the root of technological progress, is based on the contradiction that Western conditions and wealth were gained through the colonies that were pillaged and subjugated. The previously colonized

share a collective history that Meghji argues should form the basis of a decolonial standpoint (2021). A new account of modernity is required to construct a decolonial narrative, acknowledging that colonial wealth was built on exploitation and violence (Meghji 2021). This activates the agency of the colonized to produce counter-narratives to the colonial matrix of power. Decolonial studies highlight colonial continuities enabled by digital technologies (Mohamed, Png, and Isaac 2020).

Technological optimism

Technological optimism is a rationality positioning technology as the key to societal advancement. Under the umbrella of innovation, technological progress, closely linked to economic growth, is believed capable of solving large problems at scale, including problems that technology itself creates. The growth imperative fuels an acceptance of collateral damage as a necessary byproduct. As Meghji argues, coloniality has shaped much of the modern world's knowledge production (2021). This rationality informs tolerance to surveillance, the patriotic framing of technologies as national assets, responses to systemic failures, and the narrative of the successful technology entrepreneur. Digital realities are imbued with these ideas, producing information warfare, addiction, algorithmic discrimination, and the thinning of democratic institutions. The dissertation's cases represent alternatives to this rationality.

Enchanted determinism

Enchanted determinism, a term from Crawford's *Atlas of AI* (2021), describes the dual quality of AI systems as simultaneously beyond comprehension and deterministic in their capacity to uncover reliably applicable patterns. The concept arises from AI's roots in military signal processing and optimization, where the powers of rationalization and prediction were developed. The layering of abstractions gives technologies an almost mystical quality, the impression of true innovation because so few can comprehend what is

being presented. This quality also serves to shutter criticism and attempts at regulation. Enchanted determinism is built on two divergent streams: technology utopianism, where computational power solves any problem, and a dystopian view where technologies are blamed for adverse outcomes without considering the conditions of their development (Campolo and Crawford 2020).

Algorithmic inequality

Algorithmic inequality describes how mathematical models embedded in automated decision-making systems enshrine human biases in code, reinforcing oppressive social relationships. Algorithms are purposely opaque, citing intellectual property concerns, and rely on scale (O'Neil 2016). Noble (2018) demonstrates that capital, race, and gender generate inequality that algorithms embed in abstractions. Rabyniuk (2020) argues that the public should have the right to access, critique, and regulate algorithmic models. Social knowledge assembled through complex correlations and algorithmic computations creates social norms through "governance by proxy," the use of opaque data computation techniques for decision-making (Couldry and Mejias 2019). This algorithmic discrimination is most intensely experienced by the marginalized. AI systems are not objective; they are designed to classify and therefore express power emanating from the political economy (Crawford 2021).

Technical division of labor

The technical division of labor concerns how decentralization and coherent production are simultaneously managed in the technologically-infused knowledge economy. Unger argues that the answer is not more technology; evolutions in cultural and institutional arrangements are necessary, and a repressed transformative potential exists (Unger 2019). The current state of labor relations makes workers behave like machines rather than minds. Unger accuses the paradigm of insular vanguardism of mental colonialism, an internalized perception of cultural inferiority as an effect of colonization (2019). His position on the

political economy argues for imaginative capabilities, economic experiments, high-energy democratic engagement, and educational reform. A decolonizing vision is thus framed in terms of the political economy of technology-enabled knowledge production.

Digital surveillance

Ivan Manokha examines the case of Cambridge Analytica to demonstrate the emergence and tendencies of platform capital (Manokha 2018). He argues that data is Polanyi's fourth fictitious commodity. Platform capital requires continuous data collection, analysis, and monetization; therefore it depends on surveillance to acquire data. Surveillance is the DNA of platform capital. The coloniality of digital surveillance infrastructure can be inferred from Couldry and Mejias' data colonialism arguments: directing flows from colony to metropolis, the violence in the loss of autonomy and sovereignty, the subjugation inherent in dispossession, and the colonial organization of resources and knowledge that is the cloud (Couldry and Mejias 2019).

Data colonialism

Data colonialism explains the rise of global data flows in terms of the extractive and violent practices of historical colonialism. Couldry and Mejias (2019) identify data colonialism's basic features: processing of increasing data volumes, universalization of logistics as militarized management, datafication of labor, and the creation of data relations as social relations in a digital context. Systems contributing to data colonialism are built on six intertwined capacities: an infrastructure for extraction, a social order compelling participation, an economic model profiting from both, a form of governance based on these three, a rationale directing the system, and the architecture of a social knowledge structure (Couldry and Mejias 2019). Data capture is made to appear natural by propagating the rationality that data is freely available and the social contract of data for free services is fair.

Data relations

Data relations are the lifeblood of data colonialism. Couldry and Mejias describe them as a new form of human relations that give data colonialists a comprehensive view of sociality (2019). As historical colonialism operated through dispossession of land, data colonialism operates through the datafication of social life: social lives are not material wealth until they are datafied (Couldry and Mejias 2019). Data relations create a system of social knowledge under corporate control, where corporations maintain a parallel, unregulated social imaginary configured to optimize for benefit and power (Couldry and Mejias 2019). This dissertation treats data relations as a condition, shaping how cases navigate sovereignty construction across digital domains.

The ideology of connection

The ideology of connection posits the connection of people, things, and processes, mediated by technology, as natural. This enables the annexation of social life to capital. Connection is not inherently negative in social life, but in this context, connection is to a deeply exploitative infrastructure, creating conditions for subjugation (Couldry and Mejias 2019).

Social quantification

Social quantification refers to the sector of entities engaging in data colonialism by creating a basis for data relations and extracting economic value. Central to social quantification is the process of dispossession (Couldry and Mejias 2019). Gray observes that dispossession implies possession by some other entity; not being worthy of possession of data is a vital feature of data's commodification in the absence of digital rights (Gray 2021). Tracking as a requirement of daily life produces a continual state of subjugation that constitutes dispossession. Social quantification moves datafied social life to the cloud,

defining terms of data ownership through centralized data banks that claim ownership and gatekeeping rights.

Technologists as activists

An activist is a person who has decided to take part in changing society, stepping into the extraordinary work of making history (Flacks 1988). A movement is collective action towards shared goals for changing society; when this action produces a society that is more free for more people, it is described as a social movement (Touraine 1988; Buechler 2000). This dissertation positions technologists as a form of activist working for societal change, and technology as a form of movement. Technologists make technology not merely in response to short-term external conditions, nor only as a matter of shared identity, but as strategic actors thinking about how their technology could change the world (Deepak and Manski 2025). Digital technologists are the primary designers and builders of digital worlds.

Digital commoners

Digital commoners are a subset of digital technologists whose efforts aim to expand the spheres of life held in common, strengthen mutual aid, and create the conditions for shared participation in power. They wish to grow the commons, solidarity, and democracy. They create institutions online that can be managed and owned cooperatively, build digital systems for participatory planning and governance, and integrate distributed ledger technology with new architectures to replace the highly centralized internet. Their labor is critical to the constitution of future society (Deepak and Manski 2025).

Digital commonwealth

A digital commonwealth is a system where communities use digital technology to organize economic relations to produce democracy and solidarity (Manski and Manski 2018; Hanna and Lawrence 2020). A commonwealth is a form of economic polity wherein the organization of the economy is designed to shape politics (Winch 1977). It seeks holistic

social institutions with distinct political economies, building infrastructures that empower the public and expand technological sovereignty (Dulong de Rosnay and Stalder 2020). The elements of a digital commonwealth are mutually reinforcing, surfacing an economic polity enabled, governed, coordinated, and convened through digital technology (Manski 2017). Chapter 2 defines digital commonwealth scholarship through seven strands; Chapter 8 connects the concept to digital rights projects as the formations through which commonwealth is constructed.